Yitian (Grace) Lau

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Summary

Backend-focused full-stack Java Engineer with 3+ years of building distributed payment, machine learning and LLM-powered systems for the public sector.

Education

Carnegie Mellon University, *M.S. in Information Systems* ***01/2021 - 12/2022***

University of Colorado, *B.S. in Mathematics and Economics* ***09/2016 - 05/2020***

Experience

Software Engineer II, Pennsylvania Department of Transportation(Turnpike) ***01/2025 - Present***

Traffic Operations Copilot

- Built an **LLM agent** using tool calling on real-time traffic and weather APIs and **RAG** on historical work orders via pgvector to forecast incident risk and autonomously draft dispatch recommendations

Transaction Auditing & Reconciliation Platform

- Engineered Pennsylvania Turnpike's core billing infrastructure in **Java/Spring Boot** across 560+ highway miles, automating vehicle verification, transaction generation, and payment processing
- Optimized DMV lookup with **Elasticsearch** fuzzy matching and **Redis** caching, reducing latency from 400ms to 50ms and improving plate match accuracy by 3%
- Designed a distributed stream processing pipeline with **Kafka** and **Flink**, detecting payment mismatches via interval-join for automated financial auditing across $18M+/month in revenue

Software Engineer I, Pennsylvania Department of Transportation(Turnpike) ***06/2023 - 12/2024***

- Implemented an end-to-end **ML infrastructure** pipeline for license plate recognition using **computer vision** algorithms on **Linux**, addressing a 13% unpaid toll rate across 2M+/month transactions
- Built a **Python/FastAPI** data pipeline processing 67k+/day camera images, routing below-threshold results via Azure Service Bus to reviewer queues based on experience level
- Built a full-stack human-in-the-loop annotation platform in **React/TypeScript**, dockerized and deployed to Azure via DevOps CI/CD, enabling 25k+ image labels per day

Software Engineer(Intern to full-time), SAP ***07/2019 - 12/2020***

- Expanded automated test coverage with **JUnit** and **Mockito** unit and integration suites, raising line coverage from 50% to 80% across 22 Onboarding components
- Refactored state management of SuccessFactors Onboarding module using **React** and **JavaScript** with form pre-population and field mapping, eliminating manual data re-entry

Skills

Languages: Java, Python, Go, C++, JavaScript, TypeScript, HTML, CSS

Backend: Spring Boot, Django REST Framework, Node.js, Kafka, Flink, Azure Service Bus, Celery, gRPC

Frontend: Angular, React, Vue, Next.js, Tailwind CSS

Databases: PostgreSQL, MySQL, Microsoft SQL Server, MongoDB, Redis, Elasticsearch

Cloud & DevOps: Azure, AWS (ECS, RDS, S3), Docker, Kubernetes, Terraform

AI/ML: LLM API Integration, ML Infrastructure, Computer Vision, LangChain, Prompt Engineering, MCP